

Einsum, treewidth, and the convergence rates of multiplicative series

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Abstract

A surprisingly large family of problems in analysis, dynamics, combinatorics and probability ask for the exponential growth rate of a *partition function* $Z_L = \sum_I \omega(I)$, where I ranges over admissible “words” of length L and the weight ω is (sub)multiplicative along the word. The limit $P = \lim_L \frac{1}{L} \log Z_L$ is variously called the pressure, free energy, growth rate, or a critical exponent, and a recurring working method of the practitioner is to *conjecture* the rate $\lambda = e^P$ (with a possible power-law correction L^θ), compute Z_L for growing L , and check that, after dividing out the conjecture, the sequence tends to 1. Our companion paper observed that a V -statistic is exactly an EINSUM (a tensor contraction with no free index) and that its exact cost is $\Theta(n^{\text{tw}(G_{\mathcal{A}})+1})$, governed by the treewidth of a decomposition graph. We observe that the very same einsum/treewidth calculus governs the transfer-matrix engines that produce Z_L , and that the “decomposable vs. entangled” dichotomy a practitioner runs into is precisely the dichotomy between *bounded* and *growing* treewidth of the weight’s dependency graph. This yields (i) a clean dictionary between the two worlds; (ii) a precise version of the “divide out and check $\rightarrow 1$ ” heuristic, which is rigorous whenever the weight compresses into a finite transfer operator and in which the spectral gap is read off from the *speed* of convergence; (iii) a structured catalogue of examples spanning both regimes; and (iv) four worked computations that reproduce known constants (φ , the hard-square constant, $\dim_H E_2$, and an affinity dimension) and illustrate each phenomenon. The treewidth bound emerges as a complexity scale for the numerical engines that mathematicians already use to certify convergence rates.

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1 Introduction

1.1 A template that keeps reappearing

Consider a finite alphabet $[q] = \{1, \dots, q\}$, a set $\mathcal{W}_L \subseteq [q]^L$ of *admissible words* of length L , and a nonnegative weight ω defined on $\mathcal{W} := \bigcup_L \mathcal{W}_L$ that is *submultiplicative* along concatenation, $\omega(IJ) \leq \omega(I)\omega(J)$. The associated *partition function* and *pressure* are

$$Z_L = \sum_{I \in \mathcal{W}_L} \omega(I), \quad P = \lim_{L \rightarrow \infty} \frac{1}{L} \log Z_L, \quad \lambda = e^P. \quad (1)$$

The limit exists by Fekete’s lemma whenever $\mathcal{W}$ is closed under concatenation and ω is submultiplicative. Depending on the field, P is called the *topological pressure*, the *free energy*, the *growth rate*, the (top) *Lyapunov exponent*, or a *critical exponent*; λ is a connective constant, an entropy constant, a Stanley–Wilf limit, a joint spectral radius, and so on. The entire apparatus of *thermodynamic formalism* and *transfer operators* [2, 33, 36] exists precisely to compute P and its corrections.

The motivation for this note came from a concrete instance in fractal geometry. For an affine iterated function system with linear parts $T_1, \dots, T_q$, Falconer’s *affinity dimension* [16] is governed by

$$Z_L(s) = \sum_{I \in [q]^L} \varphi^s(T_{i_1} T_{i_2} \cdots T_{i_L}), \quad (2)$$

where φ^s is the singular-value function. The natural complaint is that (2) has “no decomposable structure: every factor depends on the *entire* matrix product”. The question this note grew out of is whether the einsum/treewidth toolkit of the companion paper [12] can help *compute* such series, thereby *assisting* the analytic study of their convergence and convergence rates, and – more importantly – whether (1) is a rare shape or a pervasive one. It is pervasive.

1.2 The practitioner’s heuristic

A working mathematician rarely has P in closed form. Instead one *conjectures* an asymptotic

$$Z_L \sim C \lambda^L L^\theta \quad (L \rightarrow \infty), \quad (3)$$

computes Z_L for as large an L as the budget allows, and tests the conjecture by *dividing it out*: the diagnostic ratio

$$\frac{Z_{L+1}/Z_L}{\lambda} \left(1 + \frac{1}{L}\right)^{-\theta} \longrightarrow 1 \quad \text{iff the conjecture (3) is correct.} \quad (4)$$

If the ratio settles to 1 the rate is corroborated; the *speed* at which it settles, and any systematic drift, expose the subleading exponent θ and the spectral gap. The two questions we take up are: *when is this heuristic rigorous*, and *what does it cost to push L far enough to trust it?*

1.3 The companion paper, in one paragraph

Chen et al. [12] study the exact computation of higher-order U - and V -statistics. A V -statistic with a multiplicatively decomposable kernel $h(\mathbf{x}) = \prod_k h_k(\mathbf{x}[A_k])$ is, after tensorisation $T_k(\alpha) = h_k(\mathbf{X}[\alpha])$, *exactly* an Einstein summation with no output index,

$$\mathbb{V}(h, \mathbf{X}) = \text{EINSUM}(\mathcal{T}, \mathcal{A}) = \sum_{\alpha \in [n]^m} \prod_{k=1}^K T_k(\alpha[A_k]). \quad (5)$$

Its *decomposition graph* $G_{\mathcal{A}}$ has the index set $[m]$ as vertices and an edge between two indices that co-occur in some A_k . The central complexity result is that evaluating (5) by the optimal index elimination costs

$$\Theta(|\mathcal{A}| n^{\text{tw}(G_{\mathcal{A}})+1}), \quad (6)$$

where $\text{tw}(G_{\mathcal{A}})$ is the treewidth of $G_{\mathcal{A}}$; the optimal pairwise schedules of `opt_einsum`/`torch.einsum` attain the same bound in time *and* space. The link between contraction cost and treewidth goes back to Markov and Shi [30] in the tensor-network literature and to variable elimination in graphical models [7, 27]; finding the optimal order is NP-hard [1], which is why practical libraries [19, 37] use heuristics.

1.4 Thesis and contributions

Our thesis is a single sentence:

the transfer-matrix / finite-lattice engines that mathematicians use to estimate the rate λ in (1) are themselves EINSUM contractions, and the einsum/treewidth calculus of (6) both classifies when the “divide-out” heuristic is rigorous and quantifies its cost.

Concretely:

1. **A dictionary** (Section 2) between U -statistics and pressure computations. The growth parameter is the *word length* L (the number of summation indices), not the alphabet size q (the index range); the weight’s dependency graph G_L grows with L , and the relevant quantity is the *asymptotic treewidth* $\sup_L \text{tw}(G_L)$.
2. **A structural dichotomy** (Propositions 2.1 and 2.4): Z_L is computable in time *polynomial* in L if and only if the family is *transfer-decomposable* ($\sup_L \text{tw}(G_L) < \infty$), in which case $Z_L = \langle u, T^L v \rangle$ for a fixed transfer operator T , the pressure is $\log \rho(T)$, and the divide-out diagnostic (4) converges *geometrically at the spectral-gap rate*. Otherwise $\text{tw}(G_L)$ grows and the cost is $q^{\Theta(L)}$ – the “everything in one lump” obstruction of (2).
3. **A catalogue** (Section 4) of more than a dozen example families across both regimes, each tagged with its structure, its known rate result, and the role computation plays.
4. **Four worked computations** (Section 5) that we actually ran (code in `code/`): the golden-mean shift, hard squares on a strip, the Hausdorff dimension of the continued-fraction set E_2 , and an affinity dimension. They reproduce φ , the hard-square constant $\kappa = 1.50304808\dots$, $\dim_H E_2 = 0.5312805062772\dots$ to 15 digits, and an affinity dimension, and each isolates one phenomenon from the dictionary.

Nothing here competes with the deep analytic machinery of the cited works; the contribution is a unifying *computational* lens that connects the practitioner’s heuristic, the transfer operator, and the treewidth complexity of the companion paper.

2 From a fixed einsum to a growth-indexed family

2.1 Growth-indexed einsum families

In (5) the EINSUM expression is *fixed*: m indices, each of range n . To represent a partition function (1) we let the *number of indices grow*. Write the weight of a length- L word as a tensor network: there are tensors $\mathcal{T}_L = (T_k)$ and a signature $\mathcal{A}_L = (A_k)$ with index set $\{i_1, \dots, i_L\} \cup \{\text{bounded internal legs}\}$, all index ranges equal to q (or to a fixed internal dimension d), such that

$$Z_L = \text{EINSUM}(\mathcal{T}_L, \mathcal{A}_L) = \sum_{\alpha} \prod_k T_k(\alpha[A_k]). \quad (7)$$

The decomposition graph $G_{\mathcal{A}_L}$ now has $\Theta(L)$ vertices. Applying (6) verbatim, with the role of the sample size n played by the index range q , gives the cost of computing Z_L exactly.

Proposition 2.1 (Cost of a growth-indexed family). *The exact evaluation of Z_L via (7) by optimal index elimination costs $\Theta(|\mathcal{A}_L| q^{\text{tw}(G_{\mathcal{A}_L})+1})$. In particular, writing $W_L := \text{tw}(G_{\mathcal{A}_L})$ and noting $|\mathcal{A}_L| = \Theta(L)$ for the chain-type networks below,*

- if $\sup_L W_L = W < \infty$, the cost is $\Theta(L q^{W+1})$ – polynomial (indeed linear) in the growth parameter L ;
- if $W_L = \Theta(L)$ (e.g. $G_{\mathcal{A}_L} = K_L$), the cost is $q^{\Theta(L)}$ – exponential in L .

Proof. Immediate from the index-elimination bound of Chen et al. [12, Prop. 1] applied to $\mathcal{A}_L$ with index range q , together with the matching lower bound for optimal pairwise schedules. The two bullets substitute the stated W_L . $\square$

The transfer-matrix dimension is exactly q^W up to constants: a tree decomposition of width W carries states indexed by the $\leq W+1$ live indices at a bag, i.e. q^{W+1} values, and elimination marches a separator of q^W states along the chain. This identifies the two folklore cost measures.

Principle 2.2 (Decomposable = bounded treewidth). “ ω compresses into a finite transfer matrix” $\iff$ “ $\sup_L \text{tw}(G_{\mathcal{A}_L}) < \infty$ ”. The practitioner’s feeling that a weight is *entangled* (“everything depends on everything”) is the statement that the treewidth of its dependency graph grows with L .

2.2 Transfer-decomposable families and the spectral rate

We make the bounded-treewidth case precise for the linearly ordered (pathwidth-bounded) families that cover all our examples; this is the genuine transfer-matrix setting.

Definition 2.3 (Transfer-decomposable family). The family (7) is *transfer-decomposable of width W* if there is a finite state space Σ with $|\Sigma| \leq q^W$, boundary vectors $u, v \in \mathbb{R}_{\geq 0}^{\Sigma}$, and a fixed nonnegative *transfer matrix* $T \in \mathbb{R}_{\geq 0}^{\Sigma \times \Sigma}$ such that

$$Z_L = \langle u, T^L v \rangle \quad \text{for all } L. \quad (8)$$

Subshifts of finite type, one-dimensional lattice models, strips of any two-dimensional model, and linear cocycles fed through a *linear* functional are all of this form; (8) is the einsum (7) contracted along the chain, of treewidth 1 in the index i_j ’s after the internal legs of dimension $|\Sigma|$ are absorbed.

Proposition 2.4 (Spectral rate and the gap). *Let (8) hold with T primitive (irreducible and aperiodic), Perron eigenvalue $\lambda_1 = \rho(T)$, second eigenvalue modulus $|\lambda_2| < \lambda_1$, and u, v not orthogonal to the corresponding Perron eigenvectors. Put $\delta := |\lambda_2|/\lambda_1 \in [0, 1)$. Then*

- (i) $P = \log \lambda_1$, i.e. $\lambda = \lambda_1$;

<i>U</i> -statistics / einsum world	Pressure / transfer-operator world
sample size n (index range)	alphabet / state size q (index range)
kernel order m (# indices, fixed)	word length L (# indices, $\rightarrow \infty$)
V -statistic = $\text{EINSUM}(\mathcal{T}, \mathcal{A})$, no output	partition function Z_L
decomposition graph $G_{\mathcal{A}}$	dependency graph $G_{\mathcal{A}_L}$ of the weight
treewidth $\text{tw}(G_{\mathcal{A}})$	$\log_q(\text{transfer-matrix dim.}) = W_L$
index elimination / pairwise schedule	transfer-matrix multiplication / tree DP
cost $\Theta(n^{\text{tw}+1})$	cost $\Theta(L q^{W+1})$ (decomposable)
sparsification, $U \rightarrow V$ Möbius sum	subleading corrections, boundary terms

Table 1: The dictionary between the companion paper and the pressure world. The growth parameter is the number of indices.

- (ii) $Z_L = c \lambda_1^L (1 + O(\delta^L))$ with $c = \langle u, w_R \rangle \langle w_L, v \rangle > 0$, where w_R, w_L are the right/left Perron eigenvectors normalised by $\langle w_L, w_R \rangle = 1$;
- (iii) $\frac{Z_{L+1}}{Z_L} = \lambda_1 (1 + O(\delta^L))$.

Proof. By the Perron–Frobenius theorem λ_1 is a simple eigenvalue with positive eigenvectors w_R, w_L , and the spectral decomposition reads $T^L = \lambda_1^L \Pi + R^L$, where $\Pi = w_R w_L^\top$ is the rank-one Perron projection and $R = T - \lambda_1 \Pi$ has spectral radius $|\lambda_2|$. Hence $Z_L = \langle u, T^L v \rangle = \lambda_1^L \langle u, w_R \rangle \langle w_L, v \rangle + \langle u, R^L v \rangle$, and $\|R^L\| = O(|\lambda_2|^L)$ gives $Z_L = c \lambda_1^L (1 + O(\delta^L))$ with $c > 0$ by the non-orthogonality hypothesis. Statements (i) and (iii) follow by taking logarithms and ratios. (If λ_2 sits in a Jordan block of size b , the error term gains a polynomial factor L^{b-1} but the geometric rate δ is unchanged.) $\square$

Corollary 2.5 (The divide-out heuristic is rigorous, and reads off the gap). *Under Proposition 2.4, for a conjectured rate λ^* the diagnostic $\rho_L := Z_{L+1}/Z_L$ satisfies $\rho_L/\lambda^* \rightarrow 1$ if and only if $\lambda^* = \lambda_1$, and then $\rho_L/\lambda_1 - 1 = O(\delta^L)$. Consequently the spectral gap δ is identifiable from the speed at which the divided-out ratio approaches 1: a $\log |\rho_L - \lambda_1|$ vs. L plot has slope $\log \delta$.*

Corollary 2.5 is the rigorous heart of the heuristic (4): dividing out the rate not only *certifies* λ in the transfer-decomposable regime but *measures* the spectral gap. We verify it numerically to machine precision in Section 5.1.

Remark 2.6 (Why a genuine power law L^θ signals non-decomposability). A finite primitive T produces a *pure* geometric correction, not a power law: in Proposition 2.4 there is no L^θ factor. A true subexponential factor L^θ in Z_L – such as the conjectured $c_L \sim A \mu^L L^{11/32}$ for two-dimensional self-avoiding walks – can arise only from an *infinite-dimensional* transfer operator (continuous spectrum, no gap) or from *criticality* (the gap $\delta \rightarrow 1$). The very presence of a power-law correction is therefore a diagnostic that the problem sits in the entangled column; this is consistent with every entry of Table 2 below.

2.3 The dictionary

Table 1 states the correspondence. The single most important row is the first: the companion paper’s *sample size* n plays the role of the *alphabet/state size* q , while the companion paper’s *kernel order* m plays the role of the *word length* L that tends to infinity.

3 Using computation to support a convergence-rate proof

We separate what is *rigorous* from what is *heuristic*.

Decomposable regime (rigorous). When the family is transfer-decomposable, Proposition 2.4 turns the rate into a finite linear-algebra problem: $\lambda = \rho(T)$ is computed once and for all from a fixed matrix, and the divide-out diagnostic provably converges at rate δ . There is no asymptotic guesswork: the rate is *computed*, not estimated, and certified error bounds follow from rigorous eigenvalue enclosures. At the analytic end of this regime, the transfer operator is trace-class and one uses its *spectral determinant* $\det(I - zT)$, whose Taylor coefficients are cycle sums; these converge *super-exponentially* (factorially) for real-analytic expanding systems, which is how Jenkinson and Pollicott [22] certify 100 digits of $\dim_H E_2$ and McMullen [31], Pollicott [34] compute conformal dimensions and Lyapunov exponents.

Entangled regime (heuristic, but principled). When $\text{tw}(G_{\mathcal{A}_L})$ grows there is no finite T . One then computes Z_L to the largest feasible L – where Proposition 2.1 says the cost is $q^{\Theta(L)}$, so the feasible L is set by the affordable treewidth – and applies the *ratio method* and *series analysis* (Padé and differential approximants) of statistical mechanics and enumerative combinatorics [20, 23]. This is exactly the divide-out procedure (4), now used to *fit* both λ and the power-law exponent θ . It is not rigorous, but it is frequently the *only* route to the conjectured exponent and is the standard practice behind, e.g., the connective-constant corrections for self-avoiding walks and the Stanley–Wilf limit of 1324-avoiding permutations [13].

What computation contributes to a proof. In the decomposable regime, computation *is* the proof: certifying $\lambda = \rho(T)$ is a finite, checkable calculation. In the entangled regime the honest statement is that an infinite-dimensional but *low-dimensional-state* operator often re-emerges – on a projective space or flag manifold for matrix cocycles [3, 16, 25, 32] – under a *domination/cone* condition, which is the exact hypothesis that restores a spectral gap and hence rapid, rigorous computation. The practitioner’s intuition that an “approximate factorisation into independent parts” would help is, precisely, the request for a dominated splitting.

4 A catalogue of examples

We list example families, grouped by regime. Throughout, “TM” marks the transfer-matrix dimension or state space; “tw” marks the asymptotic treewidth of $G_{\mathcal{A}_L}$.

4.1 Decomposable: bounded treewidth, rigorous and fast

1. **Subshifts of finite type / topological entropy.** $Z_L =$ number of admissible length- L words $= \mathbf{1}^\top A^L \mathbf{1}$ for the 0/1 adjacency matrix A ; $\lambda = \rho(A)$, $P =$ topological entropy. $\text{tw} = 1$. The golden-mean shift (Section 5.1) is the smallest instance.
2. **One-dimensional statistical mechanics.** $Z_L = \text{tr}(T^L)$ with the Boltzmann transfer matrix T ; free energy $= \log \rho(T)$ and correlation length $= 1/\log(\lambda_1/\lambda_2)$ – *the gap of Proposition 2.4 is a physical observable*. Ising, Potts, and the hard-hexagon model (Baxter’s exact solution [4, 5]) live here.
3. **Strips of two-dimensional models.** On a width- W strip the model is one-dimensional with $\text{TM} = q^{\Theta(W)}$ and $\text{tw} = W$: hard squares [11] (Section 5.2), dimers / perfect matchings (Kasteleyn [26]), independent sets, colourings. The per-site rate converges to the bulk constant as $W \rightarrow \infty$; Proposition 2.1 makes the strip width the treewidth dial that trades accuracy against an exponential cost.

4. **Hausdorff dimension of conformal / cookie-cutter sets.** Bowen’s formula [9]: $\dim_H =$ the root s^* of $P(s) = 0$, where $P(s)$ is the pressure of $-s \log |f'|$. Computed to extreme precision via transfer operators: the continued-fraction set E_2 (Section 5.3; Jenkinson and Pollicott [21, 22]), Julia sets and Kleinian limit sets [31].
5. **Gauss–Kuzmin–Wirsing constant** ≈ 0.3036 : the second eigenvalue of the Gauss-map transfer operator [38] – literally the spectral gap δ of Proposition 2.4 for that operator.
6. **Large-deviation rate functions for Markov additive functionals.** The scaled cumulant generating function $\Lambda(k) = \lim_L \frac{1}{L} \log \mathbb{E}[e^{kS_L}] = \log \rho(T_k)$ is the pressure of a *tilted* transfer matrix T_k [14]; the Gärtner–Ellis theorem is Proposition 2.4(i).
7. **Top Lyapunov exponent of random matrix products** $\gamma = \lim_L \frac{1}{L} \mathbb{E} \log \|T_{i_1} \cdots T_{i_L}\|$ [18]: *non*-decomposable for the norm, but Pollicott [34] computes it rigorously and rapidly via an analytic transfer operator on projective space – the dominated/positive case of Section 3. (See Section 5.4 for the linear-functional surrogate, which *is* decomposable.)

4.2 Entangled: growing treewidth, conjecture-and-verify

1. **Affinity dimension of self-affine sets** (2): the singular-value function φ^s is a *global spectral* functional of the whole product, so $G_{\mathcal{A}_L} = K_L$ and $\text{tw} = L - 1$. Rigorous theory via the matrix/sub-additive pressure and equilibrium states [3, 16, 25, 32]; see Section 5.4.
2. **ℓ^p joint spectral radius** $\rho_p = \lim_L (\sum_{|I|=L} \|T_I\|^p)^{1/(pL)}$ [24, 35]: the same K_L entanglement as affinity dimension. Sobering honest notes: deciding $\rho \leq 1$ is *undecidable* [6], and the finiteness conjecture is *false* [8]. This is the hardest column.
3. **Self-avoiding walks: connective constant.** $c_L = \#\{\text{SAW of length } L\}$, $\mu = \lim c_L^{1/L}$; on the honeycomb lattice $\mu = \sqrt{2 + \sqrt{2}}$ [15]. The self-avoidance constraint is global, so finite-lattice transfer matrices have $\text{tw} =$ strip width and one pushes L as far as the treewidth budget allows [23]; the conjectured $c_L \sim A\mu^L L^{11/32}$ in 2D is obtained by dividing out μ^L and fitting the power-law exponent [20, 28] – *exactly* the heuristic (4), and a power law as in Remark 2.6.
4. **Lattice animals / polyominoes** (growth constant ≈ 4.0626) and **pattern-avoiding permutations** (Stanley–Wilf limits, finite by Marcus and Tardos [29]; the 1324 class has only numerical bounds ≈ 11.60 [13]): global constraints, series-analysis exponents.
5. **Multifractal L^q -spectrum** $\tau(q)$ of a self-similar/self-affine measure, defined through $\sum_I \mu(Q_I)^q$ [17]: decomposable for self-similar measures (a scalar potential, $\text{tw} = 1$), entangled for genuinely self-affine ones.
6. **Mahler measure / Lehmer’s problem** [10]: a product over conjugates with a different (Diophantine) flavour; included as a boundary case where the template applies but the transfer structure is subtle.

5 Four worked computations

All numbers below were produced by the scripts in `code/` (run `python3 code/run_all.py`); each reproduces a known constant as a correctness check.

5.1 Golden-mean shift: the divide-out diagnostic reads the gap

Let Z_L be the number of binary words of length L with no two adjacent 1s, the simplest subshift of finite type, with transfer matrix $T = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, so $Z_L = F_{L+2}$ (Fibonacci) and $\text{tw} = 1$.

Family	$\text{tw}(G_{\mathcal{A}_L})$	Rigorous?	What computation gives
Subshift entropy / 1D stat-mech	$O(1)$	yes	$\rho(T)$ exactly; gap = corr. length
2D model on width- W strip	W	yes	bulk rate as $W \uparrow$; cost $q^{\Theta(W)}$
Conformal Hausdorff dim. (Bowen)	$O(1)$	yes	root of $P(s) = 0$; 100+ digits
GKW constant	$O(1)$	yes	second eigenvalue = gap
Lyapunov (proj. operator)	$O(1)^*$	yes*	rate under domination
Affinity dimension	$L - 1$	partial	pressure curve; root = dimension
ℓ^p joint spectral radius	$L - 1$	no [†]	estimates; <i>undecidable</i> in general
SAW connective constant	strip W	partial	μ and power law by ratio method
Lattice animals / permutations	strip W	no	growth constant + exponent (fit)

Table 2: Catalogue at a glance. “*”: the projective transfer operator is infinite-dimensional but has a fixed low-dimensional *state space* and a gap under domination. “†”: Blondel and Tsitsiklis [6], Bousch and Mairesse [8].

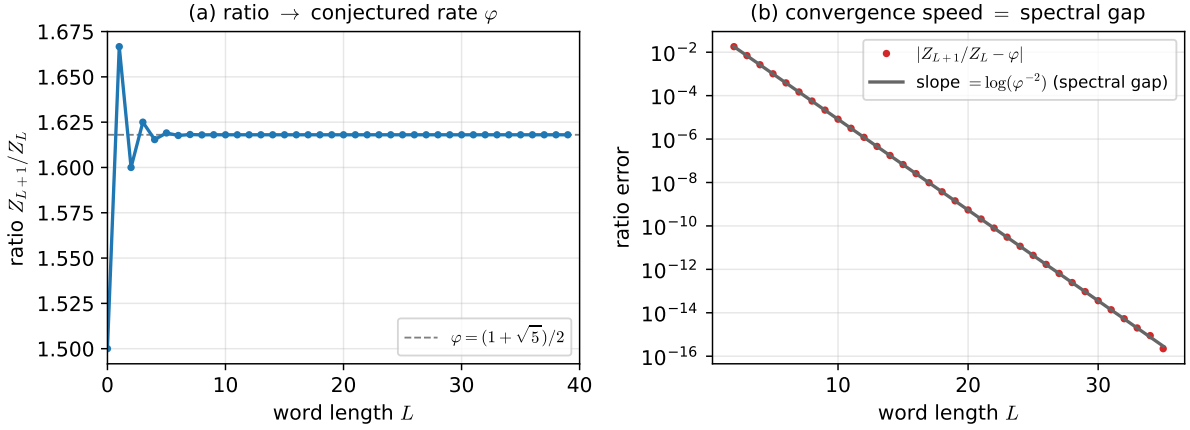


Figure 1: Golden-mean shift (decomposable, $\text{tw}= 1$). (a) $Z_{L+1}/Z_L \rightarrow \varphi$. (b) the divided-out error decays geometrically at exactly the spectral-gap rate φ^{-2} , illustrating Corollary 2.5: dividing out the rate *measures* the gap.

The eigenvalues are $\lambda_1 = \varphi = \frac{1+\sqrt{5}}{2}$ and $\lambda_2 = -1/\varphi$, so Proposition 2.4 predicts $Z_{L+1}/Z_L \rightarrow \varphi$ with geometric error at rate $\delta = |\lambda_2|/\lambda_1 = \varphi^{-2} = 0.38196601\dots$. Figure 1 confirms both: the ratio reaches $\varphi = 1.6180339887\dots$, and the divided-out error $|Z_{L+1}/Z_L - \varphi|$ falls on the line of slope $\log(\varphi^{-2})$ down to 10^{-15} . The measured per-step decay is 0.381965 against the predicted 0.381966. This is Corollary 2.5 in action: the slope of the error plot *is* the spectral gap.

5.2 Hard squares on a strip: treewidth is the accuracy/cost dial

The hard-square model (no two horizontally or vertically adjacent occupied sites) on a width- W strip is one-dimensional with transfer matrix M_W indexed by the independent sets of the path P_W ; its dimension is $F_{W+2} \sim \varphi^W$, precisely the q^{tw} of Proposition 2.1 with $\text{tw}= W$. The per-site rate $\lambda_1(M_W)^{1/W}$ and the incremental ratio $\lambda_1(M_W)/\lambda_1(M_{W-1})$ both converge to the hard-square constant $\kappa = 1.5030480824753\dots$ [5, 11], the latter *geometrically* (Figure 2): at $W = 14$ it already matches κ to 13 digits. The point is structural: wider strips give better estimates of the bulk rate, but the transfer-matrix dimension – and hence the cost, by Proposition 2.1 – grows like φ^W . *The feasible treewidth caps the attainable accuracy*, which is exactly the message the companion bound sends to the enumeration engines of Section 4.2.

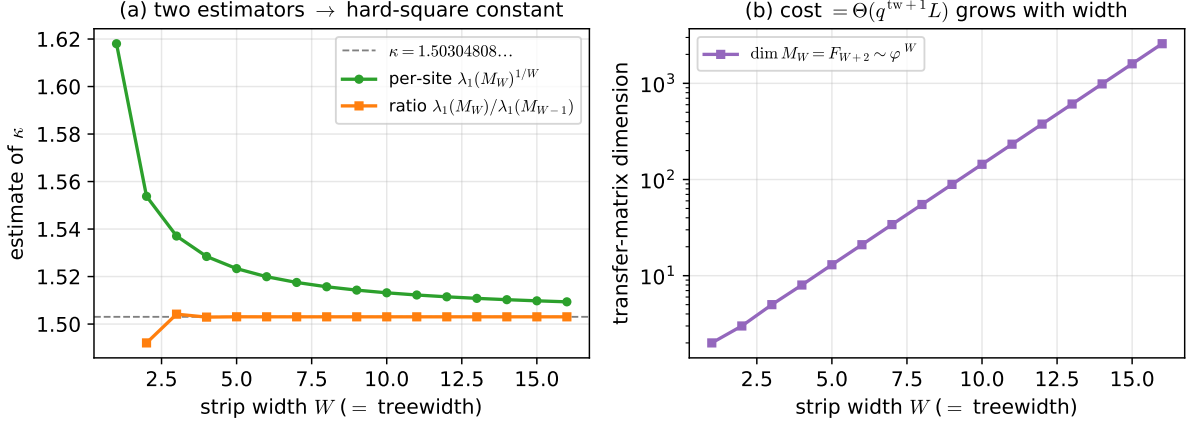


Figure 2: Hard squares on a width- W strip ($\text{tw} = W$). (a) two estimators converge to κ ; the ratio estimator is geometric. (b) the transfer-matrix dimension $= F_{W+2} \sim \varphi^W$ is the q^{tw} cost of Proposition 2.1.

5.3 Continued-fraction dimension E_2 : divide out $\rightarrow 1$ at the right s

Let E_2 be the set of continued fractions $[0; a_1, a_2, \dots]$ with all digits in $\{1, 2\}$. Its Hausdorff dimension s^* is the Bowen root of the pressure of the transfer operator $(\mathcal{L}_s f)(x) = \sum_{a \in \{1, 2\}} (a + x)^{-2s} f(1/(a + x))$, i.e. the s with leading eigenvalue $\lambda(s) = 1$. The weight $\varphi_I(x) = |D(\psi_{a_1} \circ \dots \circ \psi_{a_L})(x)|^s$ is multiplicative along the word by the chain rule, so the operator has a *fixed* finite state space even though the alphabet is infinite in the unrestricted Gauss map. Two cross-checked computations (Figure 3):

- **operator route.** A Chebyshev collocation of $\mathcal{L}_s$ on $N+1$ nodes (independent of word length) plus a root-find in s gives, at $N = 16$, $s^* = 0.5312805062772$, matching the reference value [22] to ≈ 15 digits; convergence in N is spectral (Figure 3(a)). The operator's spectral gap is $\delta \approx 0.31$.
- **word route.** Enumerating all 2^L words and forming $Z_L(s)$, the divided-out ratio $Z_{L+1}(s)/Z_L(s)$ tends to 1 *iff* $s = s^*$: at s^* it is 1.000000, while at $s^* \mp 0.03$ it locks onto 1.0387 and 0.9628 (Figure 3(b)). This is the cleanest possible realisation of “set s to the conjectured dimension and check the ratio $\rightarrow 1$ ”.

This is the prototypical case in which *computation assists a proof*: the dimension is a transcendental constant known only through such transfer-operator calculations.

5.4 Matrix products: the entangled case and when it collapses

Finally we treat the motivating family (2) on explicit 2×2 matrices. The singular-value functional couples all L indices, so the decomposition graph is K_L ($\text{tw} = L - 1$): the only exact route is brute force over q^L words. Two phenomena (Figure 4):

(a) **Collapse of the rate onto a decomposable surrogate.** Replacing the operator norm by the *linear* functional trace makes the weight multiplicative: $\sum_{|I|=L} \text{tr}(T_I) = \text{tr}(S^L)$ with $S = \sum_i T_i$, whose dependency graph is the path P_L ($\text{tw} = 1$, cost $O(L)$). For *nonnegative* matrices Perron–Frobenius forces the two pressures to coincide, $\lim_L \frac{1}{L} \log \sum_{|I|=L} \|T_I\| = \log \rho(S)$, so the cheap decomposable surrogate *proves* the rate that brute force only estimates. Numerically the surrogate gives $\log \rho(S) = 0.114986731574$ in closed form, and the brute-force norm-sum (graph K_L) drifts toward the same value (Figure 4(a)). For *signed* matrices the coincidence fails – our example gives norm-rate 0.149 against $\log \rho(S) = -0.153$ – which is the honest boundary where the trace surrogate is invalid and one needs the projective/flag-manifold operator of Section 3.

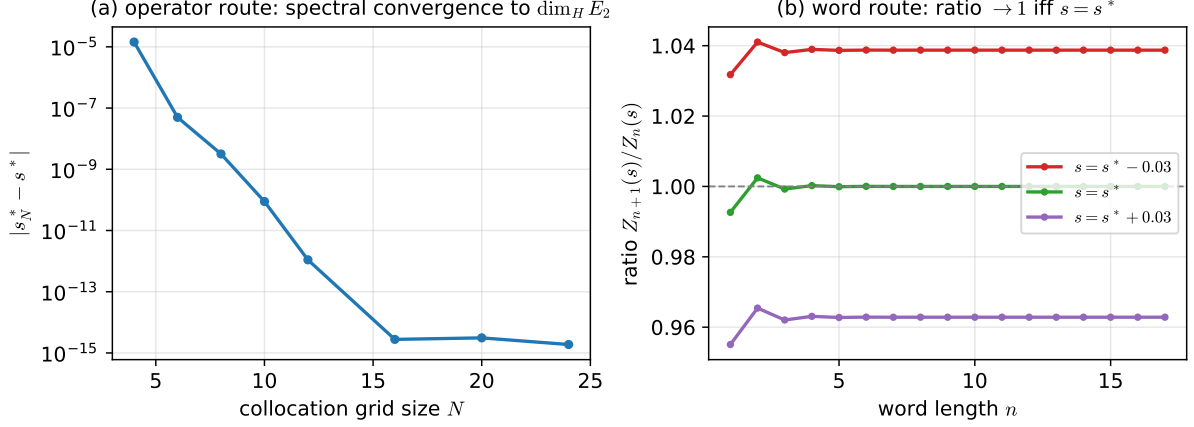


Figure 3: Hausdorff dimension of the continued-fraction set E_2 (decomposable via a fixed-state transfer operator). (a) spectral convergence of the operator estimate to $\dim_H E_2$. (b) the word-enumeration ratio $\rightarrow 1$ iff $s = s^*$.

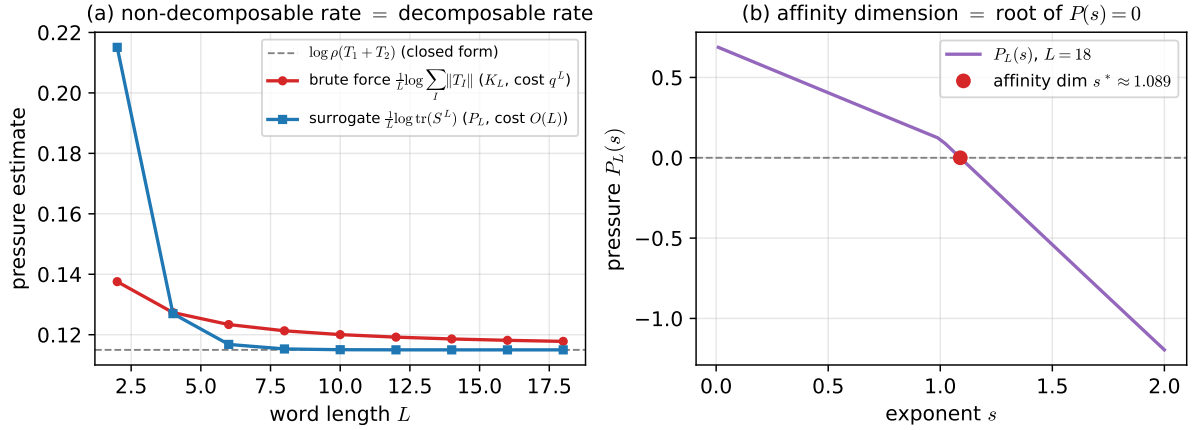


Figure 4: Matrix products (entangled, $G_{\mathcal{A}_L} = K_L$). (a) the non-decomposable norm-sum rate and the decomposable trace surrogate share the limit $\log \rho(T_1 + T_2)$ for nonnegative matrices. (b) affinity dimension as the root of the pressure $P(s) = 0$; $s = 2$ is a closed-form validation.

(b) Affinity dimension as a pressure root. With $\varphi^s = \sigma_1^s$ for $s \leq 1$ and $\sigma_1 \sigma_2^{s-1}$ for $1 \leq s \leq 2$, the affinity dimension is the root of $P(s) = 0$. The brute-force pressure curve $P_L(s)$ at $L = 18$ yields $s^* \approx 1.089$ (Figure 4(b)). The endpoint $s = 2$ is a free validation: there $\varphi^2 = |\det|$ is fully multiplicative ($\text{tw} = 0$), so $Z_L(2) = (|\det T_1| + |\det T_2|)^L$ in closed form, and the brute-force value $\frac{1}{L} \log Z_L(2) = -1.195674002$ matches the closed form to all printed digits. The decomposition $\varphi^s = |\det|^{s-1} \sigma_1^{2-s}$ ($1 \leq s \leq 2$) exhibits the split cleanly: the determinant factor is a fully decomposable scalar potential, and only the σ_1^{2-s} factor is entangled.

6 Discussion

What the treewidth theorem contributes. The catalogue's entangled column is computed, in practice, by transfer-matrix / finite-lattice engines: SAW and lattice-animal enumeration, dimer counting, hard-particle partition functions. Proposition 2.1 says these are EIN-SUM contractions whose decomposition graph is a strip of width W , so the companion bound $\Theta(q^{\text{tw}+1})$ is the complexity scale of the engines that certify convergence rates: how wide a strip (how large an L , how accurate a rate) is reachable is dictated by the treewidth, which is the

message of the hard-square experiment. The companion paper therefore speaks not only to U -statistics but to a much wider class of “numerically assisted” convergence-rate problems.

Limits. Finding the optimal elimination order is NP-hard [1]; the entangled column has genuine obstructions, including undecidability of the joint spectral radius [6] and the failure of the finiteness conjecture [8]; and the ratio method is heuristic outside the decomposable regime. The power-law diagnostic of Remark 2.6 is a useful sanity check: a clean geometric correction is the fingerprint of a finite transfer matrix.

Outlook. Three directions seem worthwhile. (i) *Off-the-shelf pressure engines*: the contraction-path optimisers `opt_einsum/cotengra` [19, 37] already solve the scheduling half of Proposition 2.1; wrapping them as a generic “ Z_L -to- P ” tool with automatic gap estimation would directly serve the practitioners of Section 4. (ii) *Automatic detection of transfer-decomposability*: deciding whether a weight admits a width- W carrier is a treewidth question on $G_{\mathcal{A}_L}$, amenable to the same heuristics. (iii) *Back to U -statistics*: degenerate or growing-order U -statistics, where the kernel order m scales with n , are themselves growth-indexed families, and the pressure viewpoint may illuminate the asymptotics of their variance and limiting distribution.

Reproducibility. The four experiments live in `code/` (`exp1–exp4`, driver `run_all.py`) and regenerate the figures in `figures/`; each prints the constant it reproduces as a self-check.

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